



The Bombay Salesian Society's

Department of Basic Science & Humanities

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(AY 2025-26)

MODULE -III

Notes Compiled By;

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MODULE 3

Geosciences, Groundwater Issues and Disaster Mitigation

3.2-I-Status of Groundwater Contamination in India (2023-24)

Source: *Central Groundwater Board (CGWB) Report*

3.2.1. The Crisis of Nitrate Contamination

While groundwater *availability* has shown slight improvement, *quality* is deteriorating due to chemical pollutants, specifically Nitrates.

- **Scale of Problem:** As of 2023, **440 districts** reported excessive nitrate levels, compared to 359 in 2017.
- **Prevalence:** **56% of India's districts** now have nitrate concentrations exceeding the safe limit of **45 mg/litre**.
- **The "Monsoon Leaching" Effect:** Contamination levels are higher during the rainy season (**32.66% of samples**) compared to pre-monsoon (**30.77%**), suggesting that rainwater carries surface pollutants into the aquifers.

3.2.2. Regional Hotspots & Other Contaminants

The report identifies specific regions where contamination has reached critical thresholds:

Contaminant	State-wise Hotspots
Nitrate	Rajasthan (49%), Karnataka (48%), Tamil Nadu (37%), Maharashtra, Telangana, Andhra Pradesh, MP.
Fluoride	Rajasthan, Haryana, Karnataka, Andhra Pradesh, Telangana.
Uranium	Rajasthan, Punjab, Haryana, Gujarat, Tamil Nadu, Andhra Pradesh, Karnataka.

3.2.3. Understanding Nitrate: Sources and Health Impacts

Nitrate contamination is primarily "anthropogenic" (caused by human activity).

A. Primary Sources

1. **Agricultural Practices:** Overuse of nitrogen-based fertilizers (Urea) and pesticides.
2. **Improper Sanitation:** Leaking septic systems and poorly managed sewage.
3. **Industrial/Waste Sites:** Leaking storage tanks and unlined landfills.

4. **Environmental Degradation: Deforestation** removes the root system that naturally filter pollutants from runoff.

B. Major Health Risks

- **Infants (High Risk):** Causes **Methemoglobinemia** (Blue Baby Syndrome). Nitrates interfere with the blood's ability to carry oxygen, leading to a bluish skin tint and potential respiratory failure.
- **Adults:** Long-term exposure is linked to thyroid dysfunction and increased risks of certain cancers.

3.2.4. Groundwater Availability & Extraction Trends

Despite the quality concerns, the physical availability of water has seen a marginal positive shift.

- **Extraction Rate:** Steady at **60.4%** across India since 2009.
- **Zonal Improvement: 73% of blocks** are now classified as '**Safe**', an increase from 67.4% in 2022.
- **Over-exploitation:** Uranium levels are notably higher in "over-exploited" zones where the water table has been pushed deep.

3.2.5. Institutional Framework: The CGWB

- **Establishment:** Founded in **1970** under the Ministry of Water Resources (now **Ministry of Jal Shakti**).
- **Apex Body:** Responsible for the management, exploration, monitoring, and regulation of groundwater.
- **Regulatory Arm:** The **Central Ground Water Authority (CGWA)**, constituted under the **Environment Protection Act, 1986**, regulates development and issues extraction permits.
- **Key Strategies:** * Implementing **Artificial Recharge** schemes.

Promoting **Rainwater Harvesting**.

Publishing District/State hydrogeological reports and Atlases.

3.2.6. Critical Terms for Revision

- **Leaching:** The process by which chemicals (like nitrates) dissolve in rainwater and seep down through the soil into groundwater.
- **Aquifer:** An underground layer of water-bearing rock or materials from which groundwater can be extracted.
- **Dark Zones:** Areas where groundwater extraction exceeds the rate of annual recharge.
- **Permissible Limit:** For Nitrates, this is **45 mg/L** (BIS/WHO standards).

3.2.7. Supplemental Context: Fluoride, Arsenic, and Aluminium

A. Fluoride (The "Double-Edged Sword")

While low levels of fluoride prevent dental cavities, levels exceeding **1.5 mg/litre** (WHO/BIS limit) lead to **Fluorosis**.

- **Geographical Spread:** Most prevalent in arid and semi-arid regions. Major hotspots include **Rajasthan, Telangana, Andhra Pradesh, and Haryana**.
- **Health Impact:**
 - Dental Fluorosis:** Permanent staining and pitting of teeth in children.
 - Skeletal Fluorosis:** Accumulation in bones causing severe joint pain, stiffness, and permanent physical deformities (crippling).
- **Source:** Weathering of fluoride-rich rocks (like Fluorite and Apatite). Over-extraction of groundwater lowers the water table, increasing the concentration of dissolved fluoride

B. Arsenic (The "Silent Killer")

Arsenic is considered one of the most toxic groundwater contaminants in India, with a safe limit of **0.01 mg/litre**.

- **Geographical Spread:** Highly concentrated in the **Ganga Brahmaputra alluvial plains**. The "Arsenic Belt" includes **West Bengal, Bihar, Uttar Pradesh, and Assam**.
- **Health Impact:**
 - Arsenicosis:** Chronic poisoning leading to skin lesions and "raindrop" pigmentation (black spots on palms/soles).

Carcinogen: Long-term exposure is a leading cause of skin, lung, and bladder cancers.

- **Source:** Naturally present in the Himalayan sediments. It is released into groundwater through complex chemical reactions, often triggered by heavy pumping for irrigation.

C. Aluminium

Aluminium has gained attention recently due to its presence in areas with acidic soil or industrial runoff. The permissible limit is **0.03 to 0.2 mg/litre**.

- **Geographical Spread:** Found in parts of **Kerala, Odisha, and West Bengal**, often where the soil is highly acidic (laterite soils).

- **Health Impact:**

Neurotoxicity: High levels are linked to neurodegenerative disorders like **Alzheimer's disease** and Parkinson's.

Bone Health: Can interfere with calcium absorption, leading to bone fragility.

- **Source:** Primarily geogenic (leaching from bauxite/laterite rocks), but significantly increased by **acid rain** and industrial waste from aluminium smelting and paper mills.

3.2.8. Comparison Table for Quick Reference

Contaminant	Permissible Limit	Primary Source	Major Health Impact
Nitrate	45 mg/L	Fertilizers/Sewage	Blue Baby Syndrome
Fluoride	1.5 mg/L	Rock Weathering	Skeletal/Dental Fluorosis
Arsenic	0.01 mg/L	Alluvial Sediments	Skin Lesions / Cancer
Aluminium	0.03 mg/L	Acidic Soil/Industry	Neurodegenerative Issues

3.2.9. Mitigation Strategies

- **Alternative Sources:** Promoting "Piped Water Supply" from surface water (rivers/dams) instead of borewells in contaminated zones.



- **Nalgonda Technique:** A specific chemical process used in India to remove Fluoride using alum and lime.
- **Arsenic Removal Units (ARUs):** Community-level filters installed at hand-pumps to chemically precipitate arsenic.
- **Dilution:** Mixing contaminated groundwater with harvested rainwater to bring chemical levels below the safety threshold.

Which one of the following ancient towns is well known for its elaborate system of water harvesting and management by building a series of dams and channelizing water into connected reservoirs? (2021)

- (a) Dholavira
- (b) Kalibangan
- (c) Rakhigarhi
- (d) Ropar

Ans: (a)

Q.2 With reference to 'Water Credit', consider the following statements: (2021)

1. It puts microfinance tools to work in the water and sanitation sector.
2. It is a global initiative launched under the aegis of the World Health Organization and the World Bank.
3. It aims to enable the poor people to meet their water needs without depending on subsidies.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans: (c)

3.2.II -Land Degradation and Desertification in India

3.2.II.1. Core Concept

- **Definition:** The decline or loss of biological/economic productivity of land caused by human activities and climate variations.
- **Scope:** Involves the deterioration of **physical** (structure), **chemical** (nutrients/salinity), and **biological** (microbial health) properties of soil.

3.2.II.2. Regional Drivers of Degradation

Region	Primary Causes	Local Impact Examples
Arid/Semi-Arid (RJ, GJ)	Overgrazing, Deforestation, Climate Change	Aravalli Hills: Dust storms and fertility loss due to tree felling.
Deccan Plateau (MH, KA, TS)	Salinization (poor drainage), Open-cast Mining	Jharia Coalfields (JH): Severe land subsidence and underground fires.
Himalayan Region	Unsustainable Tourism, Glacial Retreat	Joshimath: Land subsidence due to infrastructure pressure.

3.2.II.3. Major Consequences

1. **Food Security:** Decline in soil moisture and fertility leads to lower crop yields.
2. **Hydrological Crisis:** Reduced groundwater recharge (e.g., **Bengaluru water crisis**).
3. **Biodiversity Loss:** Habitat destruction (e.g., extinction of the **Pink-headed Duck**).
4. **Socio-Economic Shift:** Forced migration from rural zones (e.g., **Odisha**) to urban centers.



3.2.II.4. Traditional Ecological Knowledge (TEK)

Integrating indigenous wisdom is a key modern strategy for land restoration.

A. Zabo System (Nagaland)

- **The Practice:** Practiced by the **Chakhesang tribe**, *Zabo* (meaning 'to catch water') is a unique indigenous system that combines forestry, agriculture, and animal husbandry.
- **Mechanism:** Rainwater is collected from protected forest hilltops into ponds. The water then flows through cattle sheds (collecting manure/nutrients) before reaching the terraced rice fields.
- **Benefit:** Prevents soil erosion on steep slopes, maintains soil fertility naturally, and ensures water availability during dry spells.

3.2.II.B. Kuhl System (Himachal Pradesh)

- **The Practice:** A centuries-old community-managed irrigation system in the **Kangra Valley**.
 - **Mechanism:** *Kuhls* are surface channels that divert meltwater from glaciers (khuds) to village fields.
 - **Benefit:** It prevents land degradation by ensuring a controlled, equitable flow of water that prevents surface runoff and soil stripping. It relies on community cooperation (*Kohli* system) for maintenance, ensuring the land isn't over-exploited.
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3.2.II.5. Modern Sustainable Management Strategies

- **Regenerative Agriculture:** Shifting to **no-till farming** and **cover cropping** to restore the soil's organic carbon.
 - **Ecological Corridors:** Creating "green bridges" between fragmented forests to allow species movement and natural rewilding.
 - **Phytoremediation:** Planting specific species (like *Indian Mustard* or *Sunflowers*) that can naturally "suck out" heavy metals and pollutants from contaminated soils.
 - **Urban Greening:** Developing "Miyawaki Forests" and green infrastructure to reduce the heat island effect and improve soil absorption in cities.
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3.2.II.6. Key Terms for Revision

- **Salinization:** The buildup of salts in soil, often caused by over-irrigation in hot climates.
- **Rewilding:** A conservation effort aimed at restoring natural processes and wilderness areas.
- **Land Subsidence:** The gradual caving in or sinking of an area of land (common in mining and over-exploited groundwater zones).
- **Bioremediation:** Using microbes to "clean" contaminated soil.



3.2.III-Land Subsidence & Mitigation Strategies

3.2.III .1. Concept and Definition

- **Definition:** Land subsidence is the gradual settling or sudden sinking of the Earth's surface due to the removal or displacement of subsurface earth materials.
 - **The "Silent" Nature:** Unlike immediate disasters like earthquakes, it is often a gradual process occurring over decades, making it difficult to detect until infrastructure fails.
 - **Global Extent:** Estimated **12 million sq. km** of land globally is susceptible to subsidence, including major sites in the USA, China, India, Iran, and Indonesia.
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3.2.III .2. Primary Causes: Anthropogenic & Natural

A. Overexploitation of Resources

- **Groundwater & Fluid Abstraction:** Water and oil provide **Pore Pressure**—a force that helps hold up the ground. Excess withdrawal causes the underlying porous formation to compact and sink.
- **Irrigation:** Over **80%** of global water extraction is used for agriculture, making it the leading cause of large-scale subsidence.
- **Energy Extraction:** Fracking and natural gas removal (e.g., in the Netherlands) lead to reservoir compaction.

B. Mining & Subsurface Voids

- **Goaf Formation:** The extraction of solid minerals (coal/ore) leaves behind large empty spaces called **goafs**. Without proper backfilling, the "roof" eventually collapses, causing surface sinking.

C. Structural & Natural Factors

- **Infrastructure Load:** The weight of tall buildings and heavy urban masses exerts vertical pressure on soft, fine-grained deposits.
- **Natural Causes:** Volcanic/seismic activity, soil compaction, and the formation of **sinkholes** in soluble rock (limestone).
- **Soil Creep:** The slow, gravity-driven downhill movement of soil in hilly terrains.



3.2.III .3. Regional Case Study: The Himalayan & Chenab Crisis

The Himalayan foothills are uniquely vulnerable due to their "young" and unstable geology.

- **Geological Fragility:** Rocks like **gneiss** are highly weathered. During monsoons, they develop high pore pressure and lose **cohesion** (the force holding soil grains together).

- **The Joshimath Crisis (Uttarakhand):**

Known as the "sinking town," it faces unregulated construction on old landslide debris.

Mishra Committee (1976) Warnings: Noted the importance of **"Toe Support"** (not removing boulders at the hill base) and **Pucca Drainage** to prevent internal erosion.

- **Hydroelectric Projects:** Stations near vulnerable towns can alter natural water flow and destabilize slopes.

3.2.III .4. Major Impacts

- **Infrastructure Damage:** High maintenance and repair costs for roads, railways, pipelines, and building foundations.
- **Environmental Risks:** Increased flood risks, permanent reduction in aquifer storage capacity, and **saltwater intrusion** in coastal areas (e.g., Jakarta, Indonesia).
- **Water Management:** Changes in the gradient of streams and drains, disrupting natural irrigation and drainage flow.

3.2.III .5. Engineering & Management Solutions

Solution	Mechanism
Deep Soil Mixing	Injecting stabilizing agents to reinforce weak soil and prevent compression.
Artificial Recharge	Injecting water into depleted aquifers to replenish them and restore pore pressure.

Solution	Mechanism
Lightweight Construction	Using alternative materials to reduce the total load exerted on the ground.
Slope Stabilization	Planting deep-rooted trees to bind soil and installing culverts to manage drainage.
Zoning & Planning	Declaring " No-Build Zones " in high-risk areas identified by satellite technology.
InSAR Technology	Using satellite radar to track millimetric ground movements from space for early warning.

3.2.III .6. Comparison: Subsidence vs. Landslide

Feature	Land Subsidence	Landslide
Direction	Primarily Vertical (downward).	Both Vertical and Horizontal .
Geography	Can occur on flat or sloped land.	Typically occurs on steep slopes .
Speed	Usually gradual (years/decades).	Can be sudden and catastrophic.
Trigger	Subsurface resource removal.	Gravity and slope failure.

3.2.III .7. Key Revision Terms

- **Pore Pressure:** The pressure of water in soil pores that resists the weight of the overlying ground.
- **Goaf:** The void left in a mine after mineral extraction.
- **Soil Cohesion:** The internal strength of soil that keeps it from falling apart; reduced by weathering.
- **InSAR:** A satellite-based tool for monitoring ground movement.



3.2.III .8-A-Case Study: Joshimath, Uttarakhand

- **The Crisis:** Known as the "sinking town," Joshimath faced severe landslides and cracks in buildings in 2023.
 - **Factors:** High population density, unregulated construction on old landslide debris, and disruption of natural water drainage.
 - **The Mishra Committee Report (1976):** Remarkably, experts warned of this nearly 50 years ago, recommending:
 - Toe Support:** Boulders at the base of hills should not be removed as they act as natural anchors.
 - Plantation:** Extensive tree planting to bind the soil.
 - Pucca Drainage:** Preventing rainwater seepage into the ground to stop internal soil erosion.
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B- Global Examples

- **Jakarta, Indonesia:** Sinking at **25 cm/year** due to excessive groundwater pumping; the city is now being relocated to East Kalimantan.

The Netherlands: Large-scale subsidence caused by the extraction of natural gas from underground reservoirs.

C-Jharia, Jharkhand: A Case Study in Severe Land Subsidence

The **Jharia Coalfields** represent one of the most critical examples of land subsidence in India, primarily caused by nearly a century of underground coal fires and unscientific mining.

Key Features of Jharia Subsidence:

- **The Cause:** Long-standing **underground coal fires** (starting in 1916) have hollowed out the earth. As the coal burns away, it creates massive underground voids (**goafs**).
- **The Trigger:** Without the structural support of the coal seams, the overlying soil and rock layers collapse under their own weight.
- **Impact on Infrastructure:** Entire colonies, railway lines, and roads have developed deep fissures. In some cases, houses have literally been "swallowed" by the earth or collapsed due to the ground sinking.



- **The "Double Hazard":** Subsidence in Jharia is uniquely dangerous because it often opens up **fissures** that allow toxic gases and surface air to reach the fires, further fueling the combustion and causing more sinking.

Consequences:

1. **Displacement:** Thousands of families have had to be relocated to "Belgharia" (a rehabilitation township) as Jharia is declared an unsafe zone.
2. **Environmental Hazard:** Subsidence cracks allow smoke and methane to leak into the atmosphere, making the air quality in the region some of the worst in India.
3. **Economic Loss:** Crucial railway links (like the Dhanbad-Chandrapura line) have faced closures due to the unstable ground beneath the tracks.

3.4-Disaster Management:

Natural calamities of different types & intensities (Earthquakes, Hailstorms, Cyclones,

Hurricanes, Flood, Avalanches, Landslides, Tsunami, etc.) affect Nations all over the World.

Not all natural calamities can be predicted and prevented, but a state of preparedness and

ability to respond quickly to the natural calamity can considerably mitigate loss of life &

property and restore normalcy at the earliest is referred as Disaster Management.

Fundamental Concepts of Disaster Management (DM)

1. Definition

Disaster Management is a multi-disciplinary endeavor involving the organization and management of resources and responsibilities for dealing with all humanitarian aspects of emergencies. It focuses on **Preparedness, Response, and Recovery** to lessen the impact of disasters.

2. The Disaster Management Cycle

The cycle is divided into **Pre-Disaster** (Risk Reduction) and **Post-Disaster** (Recovery) phases.

Phase	Category	Key Actions
Mitigation	Pre-Disaster	Building codes, zoning, vulnerability analysis, and public education.
Risk Reduction	Pre-Disaster	Anticipatory measures to avoid future risks.
Prevention	Pre-Disaster	Engineered solutions to avoid a hazard turning into a disaster (e.g., seawalls).
Preparedness	Pre-Disaster	Early Warning Systems (EWS) , stockpiling supplies, and evacuation drills.
Response	Post-Disaster	Search and Rescue (SAR), emergency medical care, and food/water distribution.
Recovery	Post-Disaster	Build Back Better (BBB) , temporary housing, and long-term medical support.

II. The Paradigm Shift: National & Global Policy

1. Shift in Focus

There has been a global shift from **Reactive** (post-disaster relief) to **Proactive** (mitigation and preparedness) management.

- **National Policy (India, 2009 & Updates):** Mandates each Ministry/Department to allocate funds for disaster mitigation. It emphasizes that mitigation must be "built-in" to every new development project.
- **The Sendai Framework (2015-2030):** The current global roadmap focusing on "understanding disaster risk" and "investing in disaster risk reduction for resilience."
- Global framework for + * + ++

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1	Understanding disaster risk
2	Strengthening governance
3	Investing in DRR
4	Enhancing preparedness



Goals

- Reduce mortality
- Reduce economic losses
- Protect infrastructure
- Improve early warning

2. Multidisciplinary Techniques

DM integrates **Space Technology** (Satellite imaging), **Data Science** (Forecasting models), and **Structural Engineering** (Retrofitting).

III.

Earthquake Management

1. Mechanics of the Hazard

- **Hypocentre (Focus):** The actual point inside the crust where pressure is released.
- **Epicentre:** The point directly above the hypocentre on the surface.
- **Measurement:** Power is measured by **Seismometers**. Magnitude is recorded on the **Richter Scale** (Logarithmic: a jump from 5 to 6 is a 10x increase in amplitude).

Earthquake. seismic activity



2. Management Strategies

- **Engineering Standards:** All new structures must follow the **National Building Code**.
- **Soft Soil Avoidance:** Structures on uncompacted soil suffer from "Liquefaction," where the ground acts like liquid during shaking.
- **Lifeline Retrofitting:** Strengthening hospitals, power stations, and communication towers to ensure they remain functional post-quake.
- **Public Capacity Building:** Training local masons in "Earthquake-Resistant Construction" techniques (e.g., lintel bands, vertical reinforcements).

IV.

Flood Management:

1. High tide combined with a stormy weather and low atmospheric pressure is one of the

main causes of flooding in coastal regions.

2. Hurricanes bring destructive storm surges, which can quickly uplift shore water by up

to several meters. A storm surge can penetrate the beach and low-lying areas farther

inland.

3. A tsunami or tidal wave produced by an earthquake or a disturbance on the ocean

floor can generate a set of waves that slowly builds up and increases in velocity in

shallower water. Unlike a storm surge, a tsunami comes in a form of an immense wave

that, if amplified tremendously, can cause flooding and excessive damage to properties.

4. Rains: Each time there are more rains than the drainage system can take, there can

be floods. Sometimes, there is heavy rain for a very short period that result in floods.



5. Dam breaking: Dams are man-made blocks mounted to hold water flowing down from

a highland. The power in the water is used to turn propellers to generate electricity.

Sometimes, too much water held up in the dam can cause it to break and overflow the

area. Excess water can also be intentionally released from the dam to prevent it from

breaking and that can also cause floods.

6. Ice and snow-melts: In many cold regions, heavy snow over the winter usually stays

un-melted for sometime. There are also mountains that have ice on top of them.

Sometimes the ice suddenly melts when the temperature rises, resulting in massive

movement of water into places that are usually dry. This is usually called a *snowmelt flood*.

Heavy Rain / Dam Failure / Storm Surge



Excess Water



Flooding

1. Modern Causes

Beyond heavy rain, floods are increasingly caused by:

- **GLOFs (Glacial Lake Outburst Floods):** Sudden release of water from glacial lakes (common in the Himalayas).
- **Urban Heat Islands:** Increased rainfall over cities due to thermal updrafts, combined with concrete surfaces that prevent infiltration (e.g., Chennai or Bengaluru floods).
- **Dam Mismanagement:** Failure to release water gradually before peak monsoon.

2. Mitigation Framework

A. Structural Mitigation (Hard Engineering)

- **Dams & Reservoirs:** Creating "Flood Storage Space" by keeping water levels low before the monsoon starts.
- **Diversion Canals (Floodways):** Bypassing cities by channelizing excess water into dry basins or the sea.
- **Water-Gates:** Portable, self-deploying barriers that use the weight of incoming water to seal off entrances.
- **Afforestation:** Trees increase **Interception** (catching rain) and **Infiltration** (water soaking into soil), reducing surface runoff.

Method	Function
• Dams	• Store water
• Diversion canals	• Redirect water
• Water barriers	• Block flood
• Afforestation	• Reduce runoff

B. Non-Structural Mitigation (Soft Engineering)

- **Flood Plain Zoning:** Legal restrictions preventing the construction of hospitals or high-value assets in high-frequency flood zones.
- **Remote Sensing:** Using satellites to map inundated areas in real-time to guide rescue teams.
- **Forecasting & Evacuation:** Utilizing Doppler Radars and river sensors to give residents 24-48 hours of lead time.

V. Key Terms for Documentation

- **Seismic Retrofitting:** Modifying existing structures to make them more resistant to seismic activity.
- **Vulnerability:** The extent to which a community or asset is susceptible to the damaging effects of a hazard.



- **Early Warning System (EWS):** An integrated system of hazard monitoring, forecasting, and communication.
- **Building Back Better (BBB):** Using the recovery phase to increase the resilience of a community beyond its pre-disaster state.

VI. Global Outlook (Recent Examples)

- **Small Island Developing States (SIDS):** Like the Maldives, focusing on "Sea Walls" and "Artificial Islands" (e.g., Hulhumalé) to counter rising sea levels.
- **Himalayan Belt:** Focus on "Seismic Gaps" where pressure has not been released for centuries, indicating a potential mega-quake.

Retrofitting for resilience in disaster management

- Retrofitting for Resilience in Disaster Management means strengthening existing buildings and infrastructure so they can better withstand disasters like earthquakes, floods, cyclones, or fires.
- What is Retrofitting?
- Retrofitting is the process of modifying old or vulnerable structures to improve their strength, stability, and safety against hazards.
- It is a proactive approach (done before a disaster), unlike repair, which is reactive.
- Why is Retrofitting Important?
- Many buildings were built without disaster-resistant designs
- Helps reduce loss of life and property
- More cost-effective than rebuilding after disaster
- Protects critical infrastructure (hospitals, schools, bridges)

Retrofitting

Category	Details & Description
Definition	The process of modifying existing or vulnerable structures to improve their strength, stability, and safety against hazards. It is a proactive approach rather than a reactive repair.



Category	Details & Description
Primary Goal	To strengthen buildings and infrastructure (hospitals, schools, bridges) so they can better withstand earthquakes, floods, cyclones, or fires.
Importance	• Addresses buildings built without disaster-resistant designs. • Reduces loss of life and property damage. • More cost-effective than rebuilding from scratch. • Ensures continuity of critical services during emergencies.
1. Structural Retrofitting	Focus: Load-bearing elements. Methods: Adding steel braces, strengthening columns/beams, and Jacketing (wrapping in concrete or steel).
2. Seismic Retrofitting	Focus: Earthquake-prone areas. Methods: Base Isolation (separating the building from the ground), Shear Walls, and Dampers to absorb seismic vibrations.
3. Flood Retrofitting	Focus: Water-related hazards. Methods: Raising building elevations, waterproofing walls, and installing permanent or temporary flood barriers.
4. Wind/Cyclone Retrofitting	Focus: High-velocity winds. Methods: Strengthening roof connections, fixing loose components, and using wind-resistant materials.

Category	Details & Description
Wind/Cyclone Retrofitting Key Techniques	• Jacketing: Wrapping RC or steel around existing columns. • Base Isolation: Installing flexible bearings at the foundation. • Cross Bracing: Using diagonal steel supports. • FRP Wrapping: Using Fiber Reinforced Polymers for lightweight strength. • Anchor Bolting: Securing structures more firmly to their foundations.
Process Steps	1. Assessment: Identifying vulnerabilities. 2. Analysis: Calculating structural loads and weaknesses. 3. Design: Creating a technical retrofitting plan. 4. Implementation: Physical construction/modification. 5. Monitoring: Ongoing maintenance and performance checks.
Real-World Examples	• Retrofitting schools and hospitals in high-risk zones. • Large-scale strengthening projects following the 2001 Gujarat Earthquake. • Ongoing urban resilience initiatives under India's National Disaster Management programs.

Category 1: Groundwater Contamination (India 2023-24)

1. According to the 2023-24 CGWB report, what is the 'Permissible Limit' for Nitrate in drinking water?

- A) 1.5 mg/L
- B) 10 mg/L
- C) 45 mg/L
- D) 0.01 mg/L



Answer: C

2. Why are nitrate levels often higher during the monsoon season?

- A) Due to the "Monsoon Leaching" effect where rainwater carries surface pollutants into aquifers.
- B) Because nitrates are naturally produced by rain.
- C) Because industrial sites produce more waste in the rain.
- D) Due to decreased water extraction during the rainy season.

Answer: A

3. Which health condition in infants is directly linked to excessive nitrate consumption?

- A) Skeletal Fluorosis
- B) Methemoglobinemia (Blue Baby Syndrome)
- C) Arsenicosis
- D) Neurotoxicity

Answer: B

4. Which region is specifically referred to as the 'Arsenic Belt' in India?

- A) Arid regions of Rajasthan
- B) Deccan Plateau of Karnataka
- C) Ganga-Brahmaputra alluvial plains
- D) Coastal regions of Kerala

Answer: C

5. The 'Nalgonda Technique' is a chemical process specifically designed to remove:

- A) Arsenic
- B) Uranium
- C) Fluoride
- D) Aluminium

Answer: C

6. Which contaminant is linked to neurodegenerative disorders like Alzheimer's and Parkinson's?

- A) Fluoride
- B) Aluminium
- C) Nitrate
- D) Arsenic

Answer: B



Category 2: Land Degradation & Traditional Knowledge

7. What is the defining characteristic of the 'Zabo' system practiced in Nagaland?

- A) It uses glaciers to irrigate fields.
- B) It captures rainwater from hilltops and passes it through cattle sheds to enrich fields.
- C) It relies on deep-well drilling for irrigation.
- D) It is a method of urban vertical farming.

Answer: B

8. The 'Kuhl' system of Himachal Pradesh is primarily managed by:

- A) The Central Government
- B) Private corporations
- C) Local village communities (Kohli system)
- D) International NGOs

Answer: C

9. Phytoremediation involves using _____ to clean contaminated soil.

- A) Chemical injections
- B) Specific plant species (like Sunflowers)
- C) High-pressure steam
- D) Microscopic bacteria

Answer: B

10. What is 'Salinization' primarily caused by in arid regions?

- A) Acid rain
- B) Over-irrigation and poor drainage
- C) Over-extraction of coal
- D) Planting too many trees

Answer: B

Category 3: Land Subsidence (Joshimath & Jharia)

11. What is the primary cause of land subsidence in the Jharia coalfields?

- A) Excessive groundwater pumping
- B) Underground coal fires creating massive voids (goafs)
- C) Volcanic activity
- D) Excessive urban weight

Answer: B



12. In the 1976 Mishra Committee report on Joshimath, what was 'Toe Support' referring to? A) The foundation of tall buildings
B) Boulders at the base of a hill that act as natural anchors
C) A type of footwear for mountain rescue
D) The top layer of forest soil

Answer: B

13. What force helps hold up the ground but is lost when groundwater is over-extracted? A) Tectonic pressure
B) Pore Pressure
C) Atmospheric pressure
D) Centrifugal force

Answer: B

14. InSAR technology allows scientists to track ground movement using: A) Underground sensors
B) Satellite radar
C) Ocean buoys
D) Weather balloons

Answer: B

15. How does Land Subsidence differ from a Landslide?
A) Subsidence is horizontal; Landslides are vertical.
B) Subsidence is primarily vertical (downward) and usually gradual.
C) Landslides only occur on flat ground.
D) Subsidence is always sudden and catastrophic.

Answer: B

Category 4: Disaster Management & Retrofitting

16. Which phase of the Disaster Management Cycle includes 'Building Codes' and 'Vulnerability Analysis'?
A) Response
B) Mitigation
C) Recovery
D) Relief

Answer: B



17. What is the main difference between Retrofitting and Repair?

- A) Retrofitting is done after a disaster; Repair is done before.
- B) Retrofitting is proactive (strengthening before disaster); Repair is reactive.
- C) Repair is more expensive than Retrofitting.
- D) Retrofitting only applies to bridges.

Answer: B

18. 'Base Isolation' is a technique used in which type of retrofitting?

- A) Flood Retrofitting
- B) Seismic (Earthquake) Retrofitting
- C) Wind Retrofitting
- D) Fire Retrofitting

Answer: B

19. What does 'Building Back Better' (BBB) refer to in the Recovery phase?

- A) Building exactly what was there before the disaster.
- B) Increasing community resilience beyond its pre-disaster state during reconstruction.
- C) Rebuilding faster using cheaper materials.
- D) Building only in "Dark Zones."

Answer: B

20. Which global framework (2015-2030) focuses on 'Investing in Disaster Risk Reduction for Resilience'?

- A) Kyoto Protocol
- B) Paris Agreement
- C) Sendai Framework
- D) Montreal Protocol

Answer: C